

# NAME-BASED SECURE ROUTING

## Protocol Vision and Non-Negotiable Design Principles

Version 2.0

**North Star: NBSR turns a name into a secure route, not an IP address.**

Authoritative direction document for the NBSR project. The earlier feasibility study remains supporting research; this document governs protocol scope and prevents architectural drift.

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## Document status and precedence

This document defines the intended direction of Name-Based Secure Routing (NBSR). It supersedes any earlier statement that permanently limits NBSR to an enterprise overlay on top of conventional DNS. The original feasibility report remains useful for standards research, implementation options, and enterprise deployment guidance.

| Document                                    | Role                                                                                     |
|---------------------------------------------|------------------------------------------------------------------------------------------|
| NBSR Protocol Vision v2                     | Authoritative direction, invariants, scope, anti-drift rules                             |
| Name-Based Secure Routing feasibility study | Supporting technical research and implementation references                              |
| Build Week prototype                        | Proof that a name can become a temporary, policy-authorized route                        |
| Future specifications                       | Normative wire formats, state machines, interoperability profiles, and conformance tests |

## Executive summary

Name-Based Secure Routing is a proposed network protocol where applications request a destination by name and an NBSR client creates an authenticated, encrypted route or tunnel through participating NBSR infrastructure. The application does not need to discover or directly address the destination IP.

The protocol is intended to improve the foundations of Internet and enterprise connectivity by combining native naming, secure tunnel establishment, route lifecycle management, source and destination enforcement, and optional policy extensions.

**Core protocol: naming + authenticated encrypted tunnel + name-based routing + session lifecycle. Enterprise identity, device posture, compliance, and fine-grained authorization are extensions, not prerequisites for every connection.**

## 1. North Star definition

**Name-Based Secure Routing is a network protocol in which applications request a destination by name, and an NBSR client establishes an authenticated, encrypted, policy-governed route or tunnel through participating NBSR infrastructure without requiring the application to discover or directly address**

**the destination IP.**

The shortest project statement is:

NBSR turns a name into a secure route, not an IP address.

## 1.1 What NBSR changes

| Conventional model                                | NBSR target model                                                        |
|---------------------------------------------------|--------------------------------------------------------------------------|
| Resolve name to IP, then connect                  | Request name, authorize route, establish secure tunnel, then connect     |
| IP address often acts as routing and policy input | Name and cryptographic session are primary routing inputs                |
| Server is commonly publicly addressable           | Destination may remain hidden behind NBSR infrastructure                 |
| Security controls are fragmented                  | Tunnel, routing, lifecycle, logging, and optional policy are coordinated |
| DDoS filtering is mainly destination-side         | Source-side and destination-side enforcement can cooperate               |

## 2. Core scope and explicit non-goals

### 2.1 Core protocol scope

- Native name request semantics.
- NBSR client discovery and secure peer establishment.
- Mandatory encrypted and integrity-protected transport.
- Name-bound route or tunnel creation.
- Session, stream, lease, renewal, expiry, and revocation semantics.
- Gateway and regional routing selection.
- Connection migration, failover, and resumption.
- Source and destination enforcement hooks.
- Backward-compatible coexistence with conventional DNS during migration.

### 2.2 Optional extension scope

- Enterprise workload identity and user identity.
- Device posture and risk signals.
- Fine-grained authorization by method, path, tenant, or application action.
- Subscription and tenant management.
- Compliance logging and audit policy.
- Advanced abuse intelligence and federated reputation.

## 2.3 Non-goals

- Inventing new cryptography when established secure transports are sufficient.
- Requiring Kubernetes for the protocol to function.
- Replacing every existing Internet mechanism in one flag-day migration.
- Claiming that NBSR prevents application vulnerabilities, malware, insider abuse, or every form of cyberattack.
- Making one global authority the sole controller of names, identities, or routing policy.

## 3. Non-negotiable protocol invariants

| Invariant               | Required meaning                                                                                    |
|-------------------------|-----------------------------------------------------------------------------------------------------|
| Name-first              | Applications request a name, not a destination IP.                                                  |
| Mandatory security      | Every native NBSR route uses authenticated, encrypted, integrity-protected transport.               |
| No optional secure mode | Encryption is not a feature flag and cannot be disabled for native conformance.                     |
| Transport agility       | Security properties are fixed; transport profiles may evolve.                                       |
| Application simplicity  | Applications do not manage tunnel cryptography, gateway selection, or key rotation.                 |
| Native naming           | A native NBSR deployment can resolve names without conventional DNS semantics in the critical path. |
| Transitional DNS        | DNS may be used for bootstrap, migration, fallback, or coexistence.                                 |
| ISP-capable             | The architecture supports source and destination ISP enforcement.                                   |
| Enterprise-optional     | Core NBSR can function without enterprise IAM.                                                      |
| Platform-neutral        | Kubernetes is a reference deployment, not a protocol dependency.                                    |

## 4. Reference architecture



### 4.1 NBSR client

The NBSR client is the protocol endpoint used by applications. It may run on a phone, laptop, router, server, operating system, customer-premises equipment device, or embedded platform.

- Expose a small local API or socket to applications.
- Resolve NBSR names and discover regional infrastructure.
- Authenticate peers and establish secure tunnels.
- Multiplex application streams.
- Handle tunnel renewal, migration, resumption, and closure.
- Enforce local security and privacy policy.
- Maintain compatibility with application wake-up and notification systems.

### 4.2 Regional NBSR infrastructure

- Name ownership and registration.
- Gateway discovery and regional selection.
- Tunnel acceptance, relay, or termination according to the security profile.
- Name-based routing without exposing durable origin addresses to clients.
- Regional high availability and cross-region failover.
- Rate limiting, abuse controls, and revocation.
- Federation with other NBSR regions, operators, clouds, and ISPs.

### 4.3 Destination service

A destination service may accept traffic only from its NBSR edge or may participate in end-to-end service authentication. The preferred design preserves end-to-end authentication whenever gateway policy does not require payload termination.

## 5. Naming and DNS transition model

The feasibility study correctly identified DNS as a useful migration and bootstrap mechanism. However, conventional DNS is not a permanent architectural requirement for native NBSR.

| Mode               | Purpose                                               | Requirement                                                           |
|--------------------|-------------------------------------------------------|-----------------------------------------------------------------------|
| Legacy coexistence | Serve existing applications and clients               | Conventional DNS answers remain available                             |
| Bootstrap mode     | Discover an NBSR entry point or compatibility gateway | DNS, SVCB/HTTPS, anycast, ISP discovery, or configuration may be used |
| Native NBSR mode   | Resolve name and create route using NBSR semantics    | No dependency on conventional DNS in the critical route decision      |
| Federated mode     | Resolve names across operators or trust domains       | Signed ownership, delegation, and federation rules                    |

**DNS is a migration mechanism and optional bootstrap plane. Native NBSR naming is the target architecture.**

## 6. Mandatory tunnel security profile

The protocol must define security properties as mandatory invariants while allowing certified transport profiles.

| Mandatory property  | Requirement                                                                  |
|---------------------|------------------------------------------------------------------------------|
| Peer authentication | The client and NBSR infrastructure authenticate the relevant peer identities |
| Confidentiality     | Application payload and protected routing metadata are encrypted             |
| Integrity           | Modification of traffic or route control data is detected                    |
| Forward secrecy     | Compromise of long-term credentials does not expose prior sessions           |

|                      |                                                                                       |
|----------------------|---------------------------------------------------------------------------------------|
| Anti-replay          | Captured handshake or route material cannot be reused outside permitted bounds        |
| Downgrade resistance | Attackers cannot force weaker protocol versions or security profiles                  |
| Key rotation         | Long-lived tunnels rotate keys without breaking active streams                        |
| Name binding         | The secure session is cryptographically bound to the requested name and route context |

Candidate transport profiles include QUIC/TLS 1.3, Noise-based transports, HBONE, and future reviewed profiles.

**A vendor cannot claim native NBSR conformance by implementing name routing without the mandatory secure tunnel properties.**

## 7. Tunnel and stream lifecycle

### 7.1 Distinct timers

| Timer/state             | Meaning                                                                                      |
|-------------------------|----------------------------------------------------------------------------------------------|
| Idle timeout            | Tunnel closes when no valid traffic or required control activity exists                      |
| Authorization lease     | Route remains authorized for a bounded interval and may be renewed                           |
| Maximum tunnel lifetime | Hard upper bound before full re-establishment is required                                    |
| Active stream           | Download, upload, video, RPC stream, backup, or other in-progress transfer                   |
| Completion grace        | Optional bounded period for an existing stream to finish after new stream creation is denied |

### 7.2 Active transfer renewal

**An NBSR tunnel must not expire solely because its original lease elapsed while authorized streams are actively transferring data.**

1. The client opens a route and stream for the requested name.
2. Before lease expiry, the NBSR client requests renewal.
3. The infrastructure revalidates revocation, policy, quotas, and tunnel state.
4. The lease is extended without interrupting the stream.
5. Keys rotate according to time or transferred-data thresholds.
6. After the final stream closes, the idle timeout may close the tunnel.

```
while tunnel.has_active_authorized_streams():
    if lease.near_expiry():
        revalidate_policy_and_revocation()
        rotate_keys_if_required()
        renew_lease()
    close_when_idle_or_revoked()
```

### 7.3 Network migration and resumption

- The protocol should support path migration when a phone changes from Wi-Fi to mobile data.
- A resumable session must remain bound to the same name, client, and authorization context.
- Application-level resume, such as byte-range download continuation, remains a separate but complementary mechanism.

## 8. Notifications and mobile wake-up

NBSR does not need to replace APNs, FCM, or equivalent operating-system wake-up services in the first implementation.

```
Backend update
-> APNs / FCM wakes application
-> application calls local NBSR client
-> NBSR opens a temporary secure tunnel
-> application retrieves content or state
-> tunnel becomes idle and closes or remains reusable
```

A future native operating-system NBSR runtime may maintain one low-power control channel for multiple applications, but this is not required for the initial protocol.

## 9. ISP-native deployment and DDoS resistance

ISP deployment is a first-class target architecture, not an afterthought. NBSR can place enforcement close to both the source and destination.

```
Compromised client
-> source ISP NBSR edge
    - subscriber/client/tunnel identification
    - source-side rate limiting or revocation
-> federated NBSR path
-> destination ISP / NBSR edge
    - independent admission control
    - destination rate limiting
    - protected service routing
```

| Control                | Source-side value                                     | Destination-side value                          |
|------------------------|-------------------------------------------------------|-------------------------------------------------|
| Client/tunnel identity | Limits a specific device, subscriber, or tunnel       | Accepts only valid route contexts               |
| Rate limiting          | Prevents abusive traffic from aggregating unchecked   | Protects service and shared infrastructure      |
| Anti-spoofing          | Source traffic must match authenticated session state | Rejects unauthenticated or invalid sessions     |
| Revocation             | Stops a compromised client close to origin            | Blocks known compromised peers or issuers       |
| Backend concealment    | Not applicable                                        | Origin service need not be publicly addressable |

NBSR does not create bandwidth. Its advantage is that participating networks can stop abusive traffic before it aggregates and can independently enforce limits again near the destination. Non-NBSR raw traffic still requires conventional upstream filtering.

## 10. Enterprise and cloud extensions

The Build Week prototype and the feasibility study provide a strong enterprise path. These controls should be implemented as extensions above the core protocol.

| Extension        | Examples                                                                              |
|------------------|---------------------------------------------------------------------------------------|
| Identity         | SPIFFE/SPIRE, user identity, device certificates, cloud workload identity             |
| Authorization    | OPA, Cedar, method/path policy, tenant and subscription rules                         |
| Device posture   | Attestation, endpoint health, risk signals                                            |
| Audit            | Route decisions, ticket IDs, policy hashes, revocation events                         |
| Compliance       | Retention, sovereignty, lawful access controls, privacy policy                        |
| Managed services | Hosted regional clusters, enterprise support, certification, service-level guarantees |

## 11. Kubernetes reference deployment

Kubernetes is the recommended first production-like implementation environment because it simplifies regional high availability, autoscaling, rollout, observability, and failure testing. It must not leak into the wire protocol as a mandatory dependency.

### Regional NBSR Kubernetes Cluster

- Native naming / registry service
- Routing control plane
- Tunnel gateway pool
- Federation service
- Rate-limit and abuse service
- Revocation distribution
- Observability pipeline
- Optional enterprise identity and policy services

- Use multiple gateway replicas and zone-aware scheduling.
- Keep control-plane and data-plane scaling independent.
- Use GitOps and signed artifacts for deployment.
- Test regional failure, partial control-plane outage, key rotation, and rollback.
- Provide non-Kubernetes binaries or appliances using the same protocol.

## 12. Trust, privacy, and federation

- No single global authority should control all NBSR names or trust relationships.
- Name ownership and delegation must be signed and auditable.
- Federation should support multiple operators, ISPs, clouds, and private trust domains.
- Gateways should observe only the metadata required for routing and enforcement.
- Payload termination must be explicit; relay-only operation should preserve end-to-end service authentication where possible.
- Raw service-name logging should be minimized, access-controlled, and short-retention by default.

## 13. Project roadmap

| Phase                         | Primary scope                                                                  | Exit criterion                                              |
|-------------------------------|--------------------------------------------------------------------------------|-------------------------------------------------------------|
| Phase 0 - Vision lock         | Approve this document, non-goals, invariants, and terminology                  | No architectural work begins without North Star alignment   |
| Phase 1 - Core client/server  | Local NBSR client API, native name request, secure tunnel, one regional server | Application connects by name without receiving backend IP   |
| Phase 2 - Lifecycle           | Streams, lease renewal, key rotation, idle timeout, resumption                 | Large transfer survives lease renewal and network migration |
| Phase 3 - Regional Kubernetes | HA gateway pools, autoscaling,                                                 | Zone failure does not break active or                       |

|                                 |                                                                 |                                                                         |
|---------------------------------|-----------------------------------------------------------------|-------------------------------------------------------------------------|
|                                 | observability, failure handling                                 | new routes beyond SLO                                                   |
| Phase 4 - ISP lab               | Source and destination edge enforcement, per-client rate limits | Abusive client is isolated without blocking unrelated clients           |
| Phase 5 - Federation            | Multi-operator name delegation, trust, revocation, routing      | Cross-domain route works without exposing durable origin addressing     |
| Phase 6 - Enterprise extensions | SPIFFE/SPIRE, OPA, device posture, audit and compliance         | Fine-grained authorization works without changing core tunnel semantics |
| Phase 7 - Standardization       | Protocol draft, conformance suite, independent implementations  | At least two interoperable implementations pass the same tests          |

## 14. Anti-drift review checklist

Every major design, feature, or pull request should answer these questions:

7. Does the application request a name rather than a destination IP?
8. Does the route create an authenticated and encrypted tunnel or equivalent secure session?
9. Are the security properties mandatory rather than optional?
10. Can the core protocol work without enterprise IAM?
11. Can enterprise identity and policy be added without changing core naming and tunnel semantics?
12. Does the design support source-side and destination-side enforcement?
13. Can the backend remain hidden when the deployment requires it?
14. Can adoption occur gradually without a flag-day Internet migration?
15. Does the wire protocol work outside Kubernetes?
16. Is Kubernetes being used as a reference platform rather than a protocol dependency?
17. Does the change reduce complexity for applications rather than move infrastructure complexity into each app?
18. Are privacy, revocation, failure, and downgrade behavior explicitly defined?

## 15. Immediate next actions

19. Treat this document as the project North Star and place it in the repository under docs/architecture/.
20. Rename the original report to indicate that it is an enterprise and cloud-native feasibility study.
21. Create a terminology document defining name, client, regional server, edge, tunnel, route, stream, lease, and federation.
22. Write a protocol state-machine draft for discovery, handshake, route creation, renewal, revocation, migration, and closure.
23. Update the roadmap so the first new implementation target is a native NBSR client/server path, not deeper coupling to DNS.

24. Keep the Build Week prototype as the enterprise authorization proof of concept.
25. Create conformance tests before expanding the implementation.

## Appendix A - Relationship to the feasibility study

The original 19-page feasibility study remains valuable for standards-aware implementation research. Its recommendations for QUIC/TLS 1.3, SPIFFE/SPIRE, Envoy, Cilium, Istio ambient, OpenZiti, OPA, OpenTelemetry, GitOps, load testing, chaos testing, and supply-chain hardening remain relevant.

| Feasibility-study position                             | Protocol Vision v2 position                                                                     |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| DNS is the permanent public naming and bootstrap layer | DNS is optional bootstrap and migration support; native NBSR naming is the target               |
| Enterprise and cloud-native internal services first    | Valid MVP scope, but not the complete protocol scope                                            |
| Workload identity is central                           | Workload identity is a major enterprise extension; core NBSR also supports consumer and ISP use |
| Gateway/tunnel options are implementation choices      | Transport profiles are flexible, but secure-tunnel properties are mandatory                     |
| Kubernetes-heavy implementation path                   | Recommended reference deployment, not a protocol dependency                                     |
| Control-plane policy and short-lived tickets           | Retained as an enterprise and managed-service extension                                         |

## Appendix B - Terminology baseline

| Term                 | Definition                                                                                                    |
|----------------------|---------------------------------------------------------------------------------------------------------------|
| Name                 | The application-visible identifier requested through NBSR.                                                    |
| NBSR client          | The local protocol endpoint that applications use to request names and open secure routes.                    |
| NBSR edge            | An ISP, cloud, enterprise, or regional infrastructure endpoint participating in tunnel and route enforcement. |
| Regional NBSR server | The regional control and data-plane infrastructure that resolves names, selects paths, and manages tunnels.   |
| Route                | The authorized name-based connectivity decision.                                                              |
| Tunnel               | The cryptographically protected transport carrying one or more streams.                                       |
| Stream               | An individual application flow multiplexed through a tunnel.                                                  |
| Lease                | The bounded authorization period for a route or tunnel.                                                       |
| Federation           | The trust, delegation, and routing relationship between NBSR operators or domains.                            |